

Aditya Kumar

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Summary

Final year Computer Science Engineering student specializing in Core Java, Object-Oriented Design, Data Structures & Algorithms, and distributed systems. Experienced in building containerized microservices, REST-based platforms, CI/CD pipelines, and algorithmic Java applications. Strong foundation in concurrent programming, Linux, Git, and scalable system design.

Education

Chandigarh University

Aug 2022 – July 2026

Bachelor of Engineering in Computer Science and Engineering

- GPA: 7.80/10.0
- **Coursework:** Data Structures & Algorithms (Java), Object-Oriented Analysis & Design, Operating Systems (Concurrency), Software Engineering, Distributed Systems, Computer Organization and Architecture, Cloud Computing, Edge and IoT Systems, Machine Learning, Software Engineering, Cybersecurity Fundamentals

Projects

Advanced Graph Theory & Algorithmic Logic (Java)

github.com/kodi73/Algorithms-usings-Graphs [🔗](#)

- Implemented a systematic library of fundamental graph and digraph algorithms, including Dijkstra's Shortest Path, Kruskal's MST, and Topological Sort.
- Demonstrated the exact type of logical reasoning and pattern recognition required to evaluate complex LLM outputs for coherence and step-by-step accuracy.
- Focused on Instruction Following by handling multi-dimensional constraints in graph theory, mirroring the dimension discipline required in Ethara's evaluation framework.

Neural Architecture Foundations & Backpropagation (Python)

github.com/kodi73/Learning-Backpropagation-Solving-XOR-Problem [🔗](#)

- Developed a from-scratch implementation of Backpropagation to solve the XOR logic problem, demonstrating a deep understanding of model behavior and gradients.
- Conducted rigorous Data Validation to ensure model convergence, directly aligning with Ethara's core mission of curating high-quality datasets for frontier models.
- Utilized Jupyter Notebooks to provide evidence-based justifications for model performance, a key requirement for professional AI output rating.

Data Integrity & Automated Validation Systems (JavaScript)

<https://github.com/kodi73/Form-Validation> [🔗](#)

- Engineered a robust Form Validation system that enforces strict data entry rules, showcasing the precision needed for production-grade data annotation.
- Applied structured thinking to ensure all inputs met predefined criteria for Accuracy and Completeness, two of the six core evaluation dimensions.
- Bridged the gap between full-stack fundamentals and the analytical rigor required to manage structured and unstructured datasets in AI pipelines.

Hard Skills

Programming Languages: Java (Proficient), Python (Intermediate), C++ (Familiar), C (Familiar), SQL (Proficient), JavaScript (Intermediate), Bash/Shell (Intermediate)

Core concepts: Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Microservices Architecture, REST APIs, Software Development Life Cycle (SDLC)

Frameworks and Tools: Docker, Kubernetes, Git, CI/CD Pipelines, Unix/Linux, Microsoft Azure, Microsoft SQL Server, Nginx

Web Technologies: HTML, CSS, JavaScript

Soft Skills

Analytical Thinking, Problem Solving, Teamwork, Communication, Fast Learner, Adaptability, Agile Collaboration, Ownership